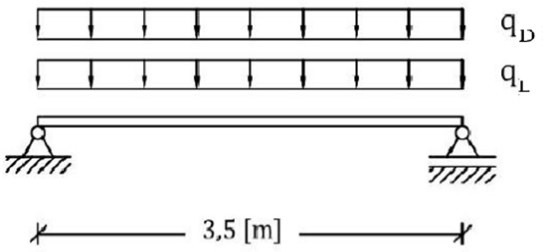
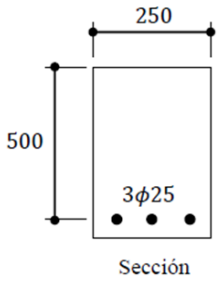
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Verifique si la viga detallada en la figura resiste las cargas solicitantes. Considere hormigón G20 y acero A630-420H. $q_D := 1000 \frac{kgf}{m}$ y $q_L := 1200 \frac{kgf}{m}$





Datos

$n_b := 3$ $\phi_b := 25\ mm$ $b := 25\ cm$ $d := 50\ cm$ $l := 3.5\ m$

$f_c := 200 \frac{kgf}{cm^2}$ $f_y := 4200 \frac{kgf}{cm^2}$ $\beta := 0.85$ $q_D := 1000 \frac{kgf}{m}$ $q_L := 1200 \frac{kgf}{m}$

$A_s := \pi \cdot \frac{\phi_b^2}{4} \cdot n_b = 14.7262\ cm^2$ $\rho := \frac{A_s}{b \cdot d} = 0.0118$

Solución

Solicitación (Mu)

$q_{u1} := 1.4 \cdot q_D = 1.4 \frac{tonnef}{m}$ $q_{u2} := 1.2 \cdot q_D + 1.6 \cdot q_L = 3.12 \frac{tonnef}{m}$

$M_{u1} := \frac{1}{8} \cdot q_{u1} \cdot l^2 = 2.1438\ tonnef\ m$ $M_{u2} := \frac{1}{8} \cdot q_{u2} \cdot l^2 = 4.7775\ tonnef\ m$

$M_u := \max([M_{u1}\ M_{u2}]) = 4.7775\ tonnef\ m$

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Forma N°1

Equilibrio de fuerzas (Suponer que As fluye)

C = T

$0.85 \cdot f_c \cdot b \cdot a = A_s \cdot f_y$

$a := \frac{A_s \cdot f_y}{0.85 \cdot f_c \cdot b} = 14.553\ \text{cm}$ $c := \frac{a}{\beta} = 17.1211\ \text{cm}$

$\varepsilon_c := 0.003$

$\varepsilon_s := \frac{\varepsilon_c}{c} \cdot (d - c) = 0.0058$ $\varepsilon_s > \varepsilon_y$ Acero en tracción fluye ==> Se verifica el supuesto

$M_n := A_s \cdot f_y \cdot \left(d - \frac{a}{2}\right) = 26.4245\ \text{tonnef m}$

Forma N°2

Equilibrio de fuerzas (Suponer que As fluye)

C = T

$0.85 \cdot f_c \cdot b \cdot a = A_s \cdot f_y$

$a := \frac{A_s \cdot f_y}{0.85 \cdot f_c \cdot b} = 14.553\ \text{cm}$ $c := \frac{a}{\beta} = 17.1211\ \text{cm}$

$c_b := d \cdot \frac{6300\ \frac{\text{kgf}}{\text{cm}^2}}{6300\ \frac{\text{kgf}}{\text{cm}^2} + f_y} = 30\ \text{cm}$ $c_b > c$ Falla dúctil ==> Acero fluye

$M_n := \rho \cdot b \cdot d^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho \cdot \frac{f_y}{f_c}\right) = 26.411\ \text{tonnef m}$

Forma N°3

Equilibrio de fuerzas (Suponer que As fluye)

C = T

$0.85 \cdot f_c \cdot b \cdot a = A_s \cdot f_y$

$a := \frac{A_s \cdot f_y}{0.85 \cdot f_c \cdot b} = 14.553\ \text{cm}$

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$$\rho_b := \frac{0.85 \cdot \beta \cdot f_c}{f_y} \cdot \frac{6300 \frac{kgf}{cm^2}}{6300 \frac{kgf}{cm^2} + f_y} = 0.0206$$

$\rho_b > \rho$ Falla dúctil ==> Acero fluye

$$M_n := \rho \cdot b \cdot d^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho \cdot \frac{f_y}{f_c}\right) = 26.411 \text{ tonnef m}$$

Resumen fuerzas

$C := 0.85 \cdot f_c \cdot b \cdot a = 61.8501$ tonnef — Fuerza de compresión en el hormigón

$T := A_s \cdot f_y = 61.8501$ tonnef — Fuerza de tracción en barras de acero

$M_n := A_s \cdot f_y \cdot \left(d - \frac{a}{2}\right) = 26.4245$ tonnef m

$M_n := 0.85 \cdot f_c \cdot a \cdot b \cdot \left(d - \frac{a}{2}\right) = 26.4245$ m tonnef

Independiente del punto de la sección que se escoja para momentar, el momento del par de fuerzas permanece constante.

$\varepsilon_s > 0.005$ Por lo tanto: $\phi := 0.9$

$\phi \cdot M_n = 23.7821$ tonnef m > $M_u = 4.7775$ tonnef m

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